

Wireless Standards — Guided Notes

Unit 2.2.1 | CompTIA Network+ N10-009 Obj. 2.2 | N10-008 2.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

The 2.4 GHz band offers better _____ and _____ than 5 GHz but has only _____ non-overlapping channels in the US and suffers from more _____ from non-Wi-Fi devices.

Question 2

The 6 GHz band, used by Wi-Fi _____ and Wi-Fi _____, provides approximately _____ MHz of spectrum. Its main limitation compared to 5 GHz is _____ and _____.

Question 3

Fill in the theoretical maximum throughput for each standard:

802.11b → _____

802.11a → _____

802.11g → _____

802.11n → _____

802.11ac → _____

802.11ax → _____

Question 4

MIMO uses multiple _____ to send parallel _____ over the same channel. A '4x4' device has _____ transmit and _____ receive antennas. If a 4x4 AP talks to a 2x2 client, the client gets _____ streams.

Question 5

MU-MIMO allows the AP to communicate with _____ clients simultaneously. Wi-Fi 5 added MU-MIMO on the _____. Wi-Fi 6 added MU-MIMO on the _____ as well.

Question 6

Wi-Fi 6 introduced _____ which lets the AP divide a channel into sub-channels to serve multiple clients per transmission. _____ reduces interference by tagging frames so devices ignore signals from neighboring networks on the same channel.

Question 7

In a high-density environment, wider channels provide _____ per client but leave _____ non-overlapping channels. Best practice for dense deployments is _____ channels with _____ AP density.

Question 8 — Real World Application

REAL WORLD: A school deploys Wi-Fi 6 APs in every classroom. During standardized testing, 30 Chromebooks per room are all active simultaneously and performance is poor. What Wi-Fi 6 features specifically address this scenario, and what configuration changes would you make?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-2-1-wireless-standards/2-2-1-wireless-standards-answer-key.pdf
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